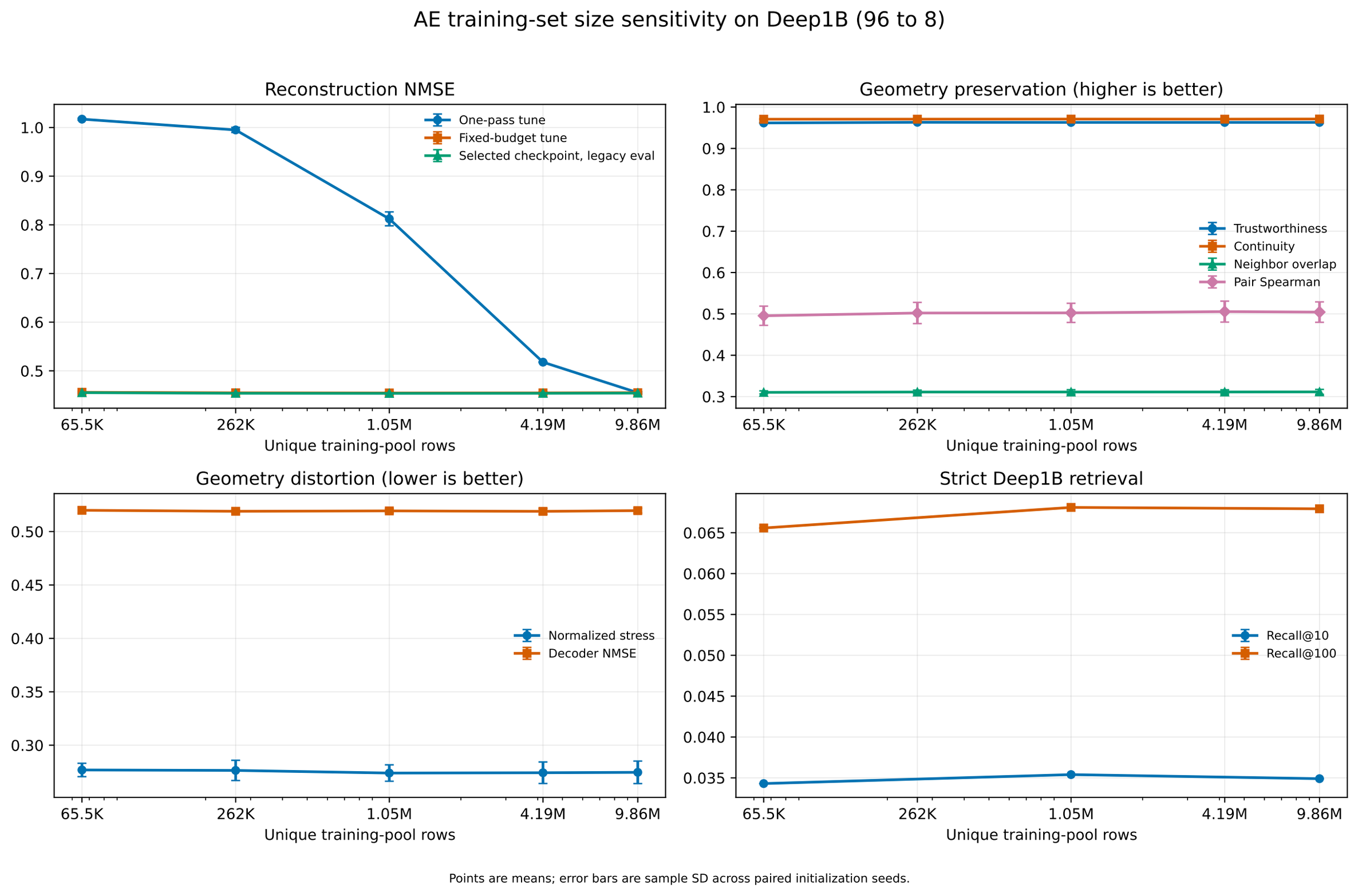
AE training-set size sensitivity on Deep1B (96 to 8)
Reconstruction NMSE
One-pass tune
Fixed-budget tune
Selected checkpoint, legacy eval
1.0
0.9
0.8
0.7
0.6
0.5
65.5K
262K
1.05M
4.19M
9.86M
Unique training-pool rows
Geometry preservation (higher is better)
Trustworthiness
Continuity
Neighbor overlap
Pair Spearman
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
65.5K
262K
1.05M
4.19M
9.86M
Unique training-pool rows
Geometry distortion (lower is better)
Normalized stress
Decoder NMSE
0.50
0.45
0.40
0.35
0.30
65.5K
262K
1.05M
4.19M
9.86M
Unique training-pool rows
Strict Deep1B retrieval
Recall@10
Recall@100
0.065
0.060
0.055
0.050
0.045
0.040
0.035
65.5K
262K
1.05M
4.19M
9.86M
Unique training-pool rows
Points are means; error bars are sample SD across paired initialization seeds.